

Title: Efficacy of follow-up after intensive care unit (ICU) discharge for improving long-term outcomes and mental health in ICU patients and informal caregivers: a systematic review and meta-analysis protocol.

Shodai Yoshihiro¹, Shunsuke Taito, PT, PhD^{2, 3}, Kota Yamauchi⁴, Shunsuke Kina⁵, Takero Terayama⁶, MD, Yusuke Tsutsumi, MD MPH DrPH^{3, 7, 8}, Yuki Kataoka, MD, MPH, DrPH^{3, 9, 10, 11} and Takeshi Unoki, RN, PhD¹².

1. Pharmaceutical department, JA General Hospital, Hiroshima, JAPAN.
2. Division of Rehabilitation, Department of Clinical Practice and Support, Hiroshima University Hospital, Hiroshima, JAPAN.
3. Systematic Review Workshop Peer Support Group (SRWS-PSG), Osaka, JAPAN.
4. Division of Rehabilitation, Steel Memorial Yawata Hospital, Fukuoka, JAPAN.
5. Division of Rehabilitation, Nakagami Hospital, Okinawa, JAPAN.
6. Department of Psychiatry, School of Medicine, National Defense Medical College, Tokorozawa, JAPAN.
7. Department of Emergency Medicine, National Hospital Organization Mito Medical Center, Ibaraki, JAPAN.
8. Department of Human Health Science, Graduate School of Medicine, Kyoto University, Kyoto, JAPAN.
9. Department of Internal Medicine, Kyoto Min-Iren Asukai Hospital, Kyoto, JAPAN.
10. Section of Clinical Epidemiology Section, Department of Community Medicine, Kyoto University Graduate School of Medicine, Kyoto, JAPAN.
11. Department of Healthcare Epidemiology, Graduate School of Medicine and Public Health, Kyoto University, Kyoto, JAPAN.
12. Department of Acute and Critical Care Nursing, School of Nursing, Sapporo City University, Sapporo, JAPAN.

Corresponding author: Shunsuke Taito, PT, PhD

Address: Kasumi 1-2-3, Minami-ku, Hiroshima, 734-8551 JAPAN
E-mail: shutaitou@gmail.com

Author contributions:

UT is the guarantor. YS drafted the manuscript.

All authors contributed to the development of the selection criteria, the risk of bias assessment strategy and data extraction criteria. YS, KY, and TY developed the search strategy. KY and TY provided statistical expertise. TT and UT provided expertise on this area. All authors read, provided feedback and approved the final manuscript.

Abstract

Introduction

Patients admitted in intensive care unit (ICU) and their informal caregivers acquired mental health disorder, which persisted after discharge. We describe a protocol for a systematic review for a question: Does follow-up after ICU discharge improve long-term outcomes and mental health compared with usual care in ICU patients and informal caregivers?

Methods and analysis

We will follow the Preferred reporting items for systematic review and meta-analysis 2020 for conducting the research. We will search the literature on MEDLINE, the Cochrane Central Register of Controlled Trials, EMBASE, the Cumulative Index to Nursing and Allied Health Literature, and APA PsycInfo systematically. We will include randomized controlled trials that assessed the effect of follow-up after ICU discharge for patients and informal caregivers. Follow-up after ICU discharge will be any service and program which include psychological monitor, care and intervention. Primary outcomes for patients and informal caregivers were depression, post-traumatic stress disorder, and all adverse events. We will assess the certainty of evidence based on the Grading of recommendations, assessment, development and evaluation approach for each patient and informal caregiver.

Ethics and dissemination

As this research is a SR/MA of literatures, ethical approval is not necessary. We will publish the results of this SR/MA in a peer-reviewed journal.

1.Introduction

Patients admitted in intensive care unit (ICU) and their informal caregivers acquired mental health disorder, which persisted after discharge.¹ Clinically important symptom of mental health disorder is depression, post-traumatic stress disorder (PTSD) and anxiety.² The prevalence of depression, PTSD, and anxiety was each about 29%³ , 34%⁴ , and 34%⁵ at 12 months from ICU discharge.

Past systematic review and meta-analysis (SR/MA) suggests that interventions for patients and their informal caregivers might improve mental health. One SR showed that ICU follow-up focused physical therapy for patients may decrease depression score and increase mental health quality of life score slightly.⁶ Another SR showed that ICU follow-up for patients may result in little to no deference for PTSD and health-related quality of life (HR-QoL)⁷. However, the SR/MA⁶ was synthesized mental outcomes of patients and family together. In addition, after published the SR/MA⁷, some RCTs for ICU follow-up of patients was published recently.^{8,9} On the other hands, there was no systematic review focused on ICU follow-up for informal caregivers. One SR/MA showed that family support conducted in the ICU was effective for reduction of ICU stay.¹⁰ One systematic review included studies of interventions conducted for family during ICU stay.¹¹

Thus, we will conduct a SR/MA of follow-up after ICU discharge for improving long-term outcomes and mental health in ICU patients and informal caregivers.

2.Research question

P: ICU patients and informal caregivers

I: Follow-up after ICU discharge

C: Usual care

O: Long-term outcomes and mental health

Does follow-up after ICU discharge improve long-term outcomes and mental health compared with usual care in ICU patients and informal caregivers?

3.Method

3.1 Protocol

We used a systematic review protocol template([dx.doi.org/10.17504/protocols.io.biqrkdv6](https://doi.org/10.17504/protocols.io.biqrkdv6)). We followed the Preferred reporting items for systematic review and meta-analysis 2020 (PRISMA-2020) for preparing this protocol.¹² We will publish this protocol in protocols.io (<https://www.protocols.io/>).

3.2 Inclusion criteria of the articles for the review

3.2.1 Type of studies

We will include randomized controlled trials that assessed the effect of follow-up after ICU discharge for patients and informal caregivers. We will not apply language or country restrictions. We will include all papers including published, unpublished articles, abstract of conference and letter.

We will exclude cluster randomized studies, quasi-randomized studies, cohort studies, case-control studies, or case series. We will not exclude studies based on the observation period or publication year.

3.2.2 Study participants

We will include ICU patients and their informal caregivers.

Patients

Inclusion criteria:

Patients after ICU discharge.

We will include patients randomized not only at ICU discharge but also at hospital discharge.

Interventions will be initiated within 1 month from hospital discharge.

Exclusion criteria:

Younger than 18 years

No consent to participate

Cognitive impairment

Informal caregivers

Inclusion criteria:

We will include informal caregivers of ICU survivors and non-survivors.

Exclusion criteria:

Less than 18 years

No consent to participate

Cognitive impairment

3.2.3 Intervention

Any service and program intervened after ICU discharge, which will include a multidisciplinary intervention, follow-up clinic, and programs. In addition to these intervention by team, intervention will be included psychological monitor, care and intervention that considers the individual patient; cognitive-behavioral therapy, counselling, letter and so on. We will incorporate any intervention period.

We will exclude the follow interventions; patient-led initiatives like ICU-diaries and ICU-records, just giving general information like web-tool and video materials, and just enhanced physical rehabilitation compared with usual care.

3.2.4 Control

Care that are commonly used for patients after ICU discharge in the country where the study was conducted, which follows existing guidelines. Usual care will be contained care by providers independent of ICU or study nurse, care by informal team, and unstructured intervention that are not programmed and protocolized. In addition, usual care will not always monitor and support mental status.

3.3 Type of outcomes

We followed core outcomes sets.^{2, 13, 14} From a medical point of view¹⁵, we selected the follow outcomes for mental health as primary outcomes.

Patients

3.3.1 Primary outcomes for patients

1. Depression

Definition: Prevalence proportion of significant symptoms of depression, based on definitions by the included study's authors.

Period: The longest period between 3 months and 1 year after randomization or ICU discharge.

2. PTSD

Definition: Prevalence proportion of significant symptoms of PTSD, based on definitions by the included study's authors.

Period: The longest period between 3 months and 1 year after randomization or ICU discharge.

3. All adverse events

Definition: Definition of adverse events are set by original authors. Incidence proportion of all adverse

events.

Period: During follow up period.

3.3.2 Secondary outcomes for patients

1. Anxiety

Definition: Prevalence proportion of significant symptoms of anxiety, based on definitions by the included study's authors.

Period: The longest period between 3 months and 1 year after randomization or ICU discharge.

2. HR-QoL

Definition: Mental component summary of the Medical Health Survey Short-Form 36 (SF-36). SF-36 was reported to use for self-reported evaluation scales for evaluation of HR-QoL.^{2, 14} If the outcome of HR-QoL will be measured by other self-reported evaluation scales, we will assess whether the scales can be synthesized with SF-36.

Period: The longest period between 3 months and 1 year after randomization or ICU discharge.

3. Pain

Definition: Self-reported evaluation scales for pain set by original authors, such as pain item within the EQ-5D-5L or SF-36, visual analogue scale for pain or other pain instruments.

Period: The longest period between 3 months and 1 year after randomization or ICU discharge.

4. Readmission

Definition: Proportion of readmission at least once.

Period: During follow up period.

5. Long term mortality

Definition: All-cause mortality of the longest period between 3 months and 1 year after randomization.

Period: The longest period between 3 months and 1 year after randomization.

Informal Caregivers

3.3.3 Primary outcomes for caregivers

1. Depression

Definition: Prevalence proportion of significant symptoms of depression, based on definitions by the included study's authors.

Period: The longest period between 3 months and 1 year after randomization or ICU discharge.

2. PTSD

Definition: Prevalence proportion of significant symptoms of PTSD, based on definitions by the included study's authors.

Period: The longest period between 3 months and 1 year after randomization or ICU discharge.

3. All adverse events

Definition: Definition of adverse events are set by original authors. Incidence proportion of all adverse events.

Period: During follow up period.

3.3.4 Secondary outcomes for caregivers

1. Anxiety

Definition: Prevalence proportion of significant symptoms of anxiety, based on definitions by the included study's authors.

Period: The longest period between 3 months and 1 year after randomization or ICU discharge.

2. HR-QoL

Definition: Mental component summary of SF-36. If the outcome of HR-QoL will be measured by

other self-reported evaluation scales, we will assess whether the scales can be synthesized with SF-36. If scales cannot be synthesized, we will list results in separate table from SF-36 or describe in the texts. Period: The longest period between 3 months and 1 year after randomization or ICU discharge.

3.4 Search method

3.4.1 Electronic search

We will search the following databases:

1. MEDLINE (PubMed);
2. the Cochrane Central Register of Controlled Trials (Cochrane Library);
3. EMBASE (Dialog);
4. the Cumulative Index to Nursing and Allied Health Literature (CINAHL) (EBSCOhost);
5. APA PsycInfo (Ovid)

See Appendix 1, 2, 3, 4 and 5 for the search strategies.

3.4.2 Other resources

We will also search the following databases for ongoing or unpublished trials:

1. the World Health Organization International Clinical Trials Platform Search Portal (ICTRP);
2. ClinicalTrials.gov;

See Appendix 6, 7 for the search strategies.

We will check the reference lists of studies, including international guidelines^{16, 17} as well as the reference lists of eligible studies and articles citing eligible studies. We will ask the authors of original studies for unpublished or additional data.

3.5 Data collection and analysis

3.5.1 Selection of the studies

Two or three independent reviewers (SY, YK and KS) will screen titles and abstracts, followed by the assessment of the eligibility based on the full texts. We will contact original authors if relevant data is missing. Disagreements between the two reviewers will be resolved by discussion, and if this fails, a third reviewer will act as an arbiter (TS). We will combine machine learning classifiers for selection process.¹⁸

3.5.2 Data extraction and management

Two or three reviewers (SY, YK and KS) will perform independent data extraction of the included studies using standardized data collection form. We will use a pre-checked form using 10 randomly selected studies.

The form will include the information on study design, study population, interventions, comparison and outcomes. Any disagreements will be resolved by discussion, and if this fails, a third reviewer will act as an arbiter (TS).

3.6 Assessment of risk of bias in included studies

Two or three reviewers (SY, YK and KS) will evaluate the risk of bias independently using the Risk of Bias 2.¹⁹ Disagreements between the two reviewers will be discussed, and if this fails, a third reviewer (TS) will be acting as an arbiter, if necessary.

3.7 Measures of treatment effects

We will pool the relative risk ratios and the 95% confidence intervals (CIs) for the following binary variables: Depression, PTSD, all adverse events, anxiety, readmission and long-term mortality.

We will pool the mean differences and the 95% CIs for the following continuous variables: HR-QoL and pain.

If several different scales have been used in the included studies, we will pool the effect estimates using standard mean differences (SMDs)

We will summarize adverse events based on the definition by the original article, but we will not perform meta-analysis.

3.8 Unit of analysis issues

Clustering at the level of the enrolled units in cluster randomised studies

In dealing with cluster-RCTs, for dichotomous data, we will apply the design effect and calculate effective sample size and number of events using the intra-cluster correlation coefficient (ICC) among each unit and the average cluster size, as described in Chapter 16.3.5 of the Cochrane Handbook.²⁰ If the ICC has not been reported, we will use the ICC of a similar study as a substitute. For continuous data, only the sample size will be reduced; means and standard deviation will remain unchanged.²⁰

Randomised cross-over studies

We will consider only data from the first period.

Multiple comparisons

All intervention groups that are relevant to this review will be included.

3.9 Handling of missing data

We will ask not-presented data to the original authors.

3.9.1 Missing outcomes

We will perform the intention-to-treat (ITT) analysis for all dichotomous data as much as possible.

For continuous data, we will not impute missing data based on the recommendation by Cochrane handbook.²⁰ We will perform meta-analysis about the available data in the original study.

3.9.2 Missing statistics

When original studies only report standard error or p-value, we will calculate the standard deviation based on the method by Altman.²¹ If we don't know these values when we contact the authors, standard deviation will be calculated by confidence interval and t-value based on the method by Cochrane handbook²⁰, or validated method.²² Validity of these methods will be analyzed by sensitivity analysis.

3.10 Assessment of heterogeneity

We will evaluate the statistical heterogeneity by visual inspection of the forest plots and calculating the I² statistic (I² values of 0% to 40%: might not be important; 30% to 60%: may represent moderate heterogeneity; 50% to 90%: may represent substantial heterogeneity; 75% to 100%: considerable heterogeneity). When there is substantial heterogeneity (I² > 50%), we will assess the reason of the heterogeneity. Cochrane Chi² test (Q-test) will be performed for I² statistic, and P value less than 0.10 will be defined as statistically significant.

3.11 Assessment of reporting bias

We will search the clinical trial registry system (ClinicalTrials.gov and ICTRP) and will perform extensive literature search for unpublished trials. To assess outcome reporting bias, we will compare the outcomes defined in trial protocols with the outcomes reported in the publications. We will assess the potential publication bias by visual inspection of the funnel plot. We will conduct Egger test to assess the publication bias. We will not conduct the test when we find less than 10 trials or trials which have similar sample size. We will also assess the potential publication bias by visual inspection of the funnel plot.

3.12 Meta-analysis

Meta-analysis will be performed using Review Manager software (RevMan 5.4.2). We will use a random-effects model. If scales of scales cannot be synthesized, we will list results in separate table or describe in the texts.

When usual care of included studies match with intervention of other, we consider with conducting network meta-analysis. We will not synthesize studies of patients and informal caregivers.

3.13 Subgroup analysis

To elucidate the influence of effect modifiers on results, we will evaluate the subgroup analyses of the primary outcomes on the following factors when sufficient data are available.

1. (For participants)

Patients

Not planned.

Informal caregivers

Informal caregivers who have lost patients or not.

2. (For intervention)

Patients

Setting of intervention (In-hospital, out-hospital, or in- and out-hospital).

Informal caregivers

Not planned.

3.14 Sensitivity analysis

We will undertake the following sensitivity analyses for the primary outcomes to assess whether the results of the review are robust to the decisions made during the review process.

1. Exclusion of studies using imputed statistics.

2. Only the participants who complete the study with complete data

3. Studies which significant symptoms of depression and PTSD was evaluated by high quality. We will select studies to measured depression by depression subscale of the Hospital-Anxiety-and-Depression-Scale³ and PTSD by The Impact-of-Event-Scale-Revised⁴.

4. Summary of findings table

Two or three reviewers (SY and TU) will evaluate the certainty of evidence based on the GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach.²³ Disagreements between the two reviewers will be discussed, and if this fails, a third reviewer (KY) will be acting as an arbiter, if necessary.

Summary of findings table will be made for the following outcome based on the Cochrane handbook.²⁰³

Patients

1. Depression

2. PTSD

3. All adverse events

4. Anxiety

5. HR-QoL

6. Pain

7. Readmission

Informal caregivers

1. Depression

2. PTSD

3. All adverse events

4. Anxiety

5. HR-QoL

5. Conflict of Interest

The authors declare no conflicts of interests.

6. Support

Self-funding.

References

1. Needham DM, Davidson J, Cohen H, Hopkins RO, Weinert C, Wunsch H, et al. Improving long-term outcomes after discharge from intensive care unit: report from a stakeholders' conference. *Crit Care Med*. 2012;40(2):502-9.
2. Needham DM, Sepulveda KA, Dinglas VD, Chessare CM, Friedman LA, Bingham CO, 3rd, et al. Core Outcome Measures for Clinical Research in Acute Respiratory Failure Survivors. An International Modified Delphi Consensus Study. *Am J Respir Crit Care Med*. 2017;196(9):1122-30.
3. Rabiee A, Nikayin S, Hashem MD, Huang M, Dinglas VD, Bienvenu OJ, et al. Depressive Symptoms After Critical Illness: A Systematic Review and Meta-Analysis. *Crit Care Med*. 2016;44(9):1744-53.
4. Parker AM, Sricharoenchai T, Raparla S, Schneck KW, Bienvenu OJ, Needham DM. Posttraumatic stress disorder in critical illness survivors: a metaanalysis. *Crit Care Med*. 2015;43(5):1121-9.
5. Nikayin S, Rabiee A, Hashem MD, Huang M, Bienvenu OJ, Turnbull AE, et al. Anxiety symptoms in survivors of critical illness: a systematic review and meta-analysis. *Gen Hosp Psychiatry*. 2016;43:23-9.
6. Rosa RG, Ferreira GE, Viola TW, Robinson CC, Kochhann R, Berto PP, et al. Effects of post-ICU follow-up on subject outcomes: A systematic review and meta-analysis. *J Crit Care*. 2019;52:115-25.
7. Schofield-Robinson OJ, Lewis SR, Smith AF, McPeake J, Alderson P. Follow-up services for improving long-term outcomes in intensive care unit (ICU) survivors. *Cochrane Database Syst Rev*. 2018;11(11):Cd012701.
8. Bloom SL, Stollings JL, Kirkpatrick O, Wang L, Byrne DW, Sevin CM, et al. Randomized Clinical Trial of an ICU Recovery Pilot Program for Survivors of Critical Illness. *Crit Care Med*. 2019;47(10):1337-45.
9. Ali Abdelhamid Y, Phillips LK, White MG, Presneill J, Horowitz M, Deane AM. Survivors of Intensive Care With Type 2 Diabetes and the Effect of Shared-Care Follow-Up Clinics: The SWEET-AS Randomized Controlled Pilot Study. *Chest*. 2021;159(1):174-85.
10. Lee HW, Park Y, Jang EJ, Lee YJ. Intensive care unit length of stay is reduced by protocolized family support intervention: a systematic review and meta-analysis. *Intensive Care Med*. 2019;45(8):1072-81.
11. Zante B, Camenisch SA, Schefold JC. Interventions in Post-Intensive Care Syndrome-Family: A Systematic Literature Review. *Crit Care Med*. 2020;48(9):e835-e40.
12. Shamseer L, Moher D, Clarke M, Gherzi D, Liberati A, Petticrew M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *Bmj*. 2015;350:g7647.
13. Major ME, Kwakman R, Kho ME, Connolly B, McWilliams D, Denehy L, et al. Surviving critical illness: what is next? An expert consensus statement on physical rehabilitation after hospital discharge. *Crit Care*. 2016;20(1):354.
14. Angus DC, Carlet J. Surviving intensive care: a report from the 2002 Brussels Roundtable. *Intensive Care Med*. 2003;29(3):368-77.
15. Marra A, Pandharipande PP, Girard TD, Patel MB, Hughes CG, Jackson JC, et al. Co-Occurrence of Post-Intensive Care Syndrome Problems Among 406 Survivors of Critical Illness. *Crit Care Med*. 2018;46(9):1393-401.
16. Excellence TNIfHaC. Rehabilitation after critical illness in adults Clinical guideline [CG83] 2009 [Available from: <https://www.nice.org.uk/guidance/cg83>].
17. Nolan JP, Sandroni C, Böttiger BW, Cariou A, Cronberg T, Friberg H, et al. European Resuscitation Council and European Society of Intensive Care Medicine guidelines 2021: post-resuscitation care. *Intensive Care Med*. 2021;47(4):369-421.
18. Marshall IJ, Noel-Storr A, Kuiper J, Thomas J, Wallace BC. Machine learning for identifying Randomized Controlled Trials: An evaluation and practitioner's guide. *Res Synth Methods*. 2018;9(4):602-14.
19. Sterne JAC, Savović J, Page MJ, Elbers RG, Blencowe NS, Boutron I, et al. RoB 2: a revised tool for assessing risk of bias in randomised trials. *Bmj*. 2019;366:l4898.
20. Higgins JPT, Thomas J, Chandler J, Cumpston M, Li T, MJ P. *Cochrane Handbook for*

- Systematic Reviews of Interventions version 6.0 (updated July 2019). (editors) WV, editor2019.
21. Furukawa TA, Barbui C, Cipriani A, Brambilla P, Watanabe N. Imputing missing standard deviations in meta-analyses can provide accurate results. *J Clin Epidemiol.* 2006;59(1):7-10.
 22. Higgins JP, White IR, Wood AM. Imputation methods for missing outcome data in meta-analysis of clinical trials. *Clin Trials.* 2008;5(3):225-39.
 23. Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol.* 2011;64(4):383-94.

Appendix 1: MEDLINE (PubMed) search strategy

1. "critical care"[MeSH Terms]
2. "critical illness"[MeSH Terms]
3. "intensive care units"[MeSH Terms]
4. 1 OR 2 OR 3
5. "critical care"[Title/Abstract]
6. "critically ill"[Title/Abstract]
7. "critical illnesses"[Title/Abstract]
8. "critical ill"[Title/Abstract]
9. "critical illness"[Title/Abstract]
10. "intensive care unit"[Title/Abstract]
11. 5 OR 6 OR 7 OR 8 OR 9 OR 10
12. 4 OR 11
13. "depressive disorder"[MeSH Terms]
14. "caregiver burden"[MeSH Terms]
15. "patient readmission"[MeSH Terms]
16. "chronic pain"[MeSH Terms]
17. "pain, intractable"[MeSH Terms]
18. "quality of life"[MeSH Terms]
19. "stress disorders, post traumatic"[MeSH Terms]
20. "anxiety disorders"[MeSH Terms]
21. 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19 OR 20
22. "readmission"[Title/Abstract]
23. "subsequent hospital admission"[Title/Abstract]
24. "readmitted"[Title/Abstract]
25. "readmit"[Title/Abstract]
26. "patient readmission"[Title/Abstract]
27. "depressive disorder"[Title/Abstract]
28. "caregiver burden"[Title/Abstract]
29. "QoL"[Title/Abstract]
30. "quality of life"[Title/Abstract]
31. "chronic pain"[Title/Abstract]
32. "intractable pain"[Title/Abstract]
33. "PTSD"[Title/Abstract]
34. "post traumatic stress disorders"[Title/Abstract]
35. "anxiety disorders"[Title/Abstract]
36. 22 OR 23 OR 24 OR 25 OR 26 OR 27 OR 28 OR 29 OR 30 OR 31 OR 32 OR 33 OR 34 OR 35
37. 21 OR 36
38. 12 AND 37
39. ("randomized controlled trial"[Publication Type] OR "controlled clinical trial"[Publication Type] OR "randomized"[Title/Abstract] OR "drug therapy"[MeSH Subheading] OR "placebo"[Title/Abstract] OR "randomly"[Title/Abstract] OR "trial"[Title/Abstract] OR "groups"[Title/Abstract]) NOT ("animals"[MeSH Terms] NOT "humans"[MeSH Terms])
40. 38 AND 39

Appendix 2: CENTRAL (Cochrane Library) search strategy

1. [mh "critical care"] OR [mh "critical illness"] OR [mh "intensive care units"]
2. critical care:ti,ab OR "critically ill":ti,ab OR "critical illnesses":ti,ab OR "critical ill":ti,ab OR "critical illness":ti,ab OR "intensive care unit":ti,ab
3. 1 OR 2
4. [mh "depressive disorder"] OR [mh "caregiver burden"] OR [mh "patient readmission"] OR [mh

- "chronic pain"] OR [mh "pain, intractable"] OR [mh "quality of life"]
5. [mh "anxiety disorders"]
6. [mh "Stress Disorders, Post-Traumatic"]
7. 4 OR 5 OR 6
8. (readmission:ti,ab OR "subsequent hospital admission":ti,ab OR readmitted:ti,ab OR readmit:ti,ab OR "patient readmission":ti,ab OR "depressive disorder":ti,ab OR "caregiver burden":ti,ab OR QoL:ti,ab OR "quality of life":ti,ab OR "chronic pain":ti,ab OR "intractable pain":ti,ab OR PTSD:ti,ab OR "post traumatic stress disorders":ti,ab OR "anxiety disorders":ti,ab)
9. 7 OR 8
10. 3 AND 9

Appendix 3: EMBASE (Dialog) search strategy

1. EMB.EXACT.EXPLODE("intensive care unit")
2. EMB.EXACT.EXPLODE("critical illness")
3. 1 OR 2
4. ab(critical care) OR ti(critical care)
5. ab(critically ill) OR ti(critically ill)
6. ab(critical illnesses) OR ti(critical illnesses)
7. ab(critical ill) OR ti(critical ill)
8. ab(critical illness) OR ti(critical illness)
9. ab(Intensive care unit) OR ti(Intensive care unit)
10. ab(ICU) OR ti(ICU)
11. 4 OR 5 OR 6 OR 7 OR 8 OR 9 OR 10
12. 3 OR 11
13. EMB.EXACT.EXPLODE("depression")
14. EMB.EXACT.EXPLODE("caregiver burden")
15. EMB.EXACT.EXPLODE("hospital readmission")
16. EMB.EXACT.EXPLODE("chronic pain")
17. EMB.EXACT.EXPLODE("intractable pain")
18. EMB.EXACT.EXPLODE("quality of life assessment")
19. EMB.EXACT.EXPLODE("anxiety disorder")
20. 13 OR 14 OR 15 OR 16 OR 17 OR 18 OR 19
21. ab(readmission) OR ti(readmission)
22. ab(subsequent hospital admission) OR ti(subsequent hospital admission)
23. ab(readmitted) OR ti(readmitted)
24. ab(readmit) OR ti(readmit)
25. ab(patient readmission) OR ti(patient readmission)
26. ab(depressive disorder) OR ti(depressive disorder)
27. ab(caregiver burden) OR ti(caregiver burden)
28. ab(QoL) OR ti(QoL)
29. ab(quality of life) OR ti(quality of life)
30. ab(chronic pain) OR ti(chronic pain)
31. ab(intractable pain) OR ti(intractable pain)
32. ab(PTSD) OR ti(PTSD)
33. ab(post traumatic stress disorders) OR ti(post traumatic stress disorders)
34. 21 OR 22 OR 23 OR 24 OR 25 OR 26 OR 27 OR 28 OR 29 OR 30 OR 31 OR 32 OR 33
35. 20 OR 34
36. 12 AND 35
37. (ab(random*) OR ti(random*)) OR (ab(placebo*) OR ti(placebo*)) OR (ab(double NEAR/1 blind*) OR ti(double NEAR/1 blind*))
38. 37 AND 36

Appendix 4: CINAHL search strategy

1. ((MH "critical care+") OR (MH "critical illness+") OR (MH "intensive care units+"))
2. ((TI "critical care" OR AB "critical care") OR (TI "critically ill" OR AB "critically ill") OR (TI "critical illnesses" OR AB "critical illnesses") OR (TI "critical ill" OR AB "critical ill") OR (TI "critical illness" OR AB "critical illness") OR (TI "intensive care unit" OR AB "intensive care unit")))
3. 1 OR 2
4. ((MH "depressive disorder+") OR (MH "caregiver burden+") OR (MH "patient readmission+") OR (MH "chronic pain+") OR (MH "pain, intractable+") OR (MH "quality of life+") OR (MH "stress disorders, post traumatic+") OR (MH "anxiety disorders+"))
5. ((TI readmission OR AB readmission) OR (TI "subsequent hospital admission" OR AB "subsequent hospital admission") OR (TI readmitted OR AB readmitted) OR (TI readmit OR AB readmit) OR (TI "patient readmission" OR AB "patient readmission") OR (TI "depressive disorder" OR AB "depressive disorder") OR (TI "caregiver burden" OR AB "caregiver burden") OR (TI QoL OR AB QoL) OR (TI "quality of life" OR AB "quality of life") OR (TI "chronic pain" OR AB "chronic pain") OR (TI "intractable pain" OR AB "intractable pain") OR (TI PTSD OR AB PTSD) OR (TI "post traumatic stress disorders" OR AB "post traumatic stress disorders") OR (TI "anxiety disorders" OR AB "anxiety disorders")))
6. 4 OR 5
7. (((MH randomized controlled trials) OR (MH double-blind studies) OR (MH single-blind studies) OR (MH random assignment) OR (MH pretest-posttest design) OR (MH cluster sample) OR (TI (randomised OR randomized)) OR(AB (random*)) OR (TI (trial)) OR (MH (sample size) AND AB (assigned OR allocated OR control)) OR (MH (placebos)) OR(PT (randomized controlled trial)) OR (AB (control W5 group)) OR (MH (crossover design) OR MH (comparative studies))))NOT (((MH animals+) OR(MH (animal studies)) OR (TI (animal model*))) NOT (MH (human))))
8. 3 AND 6 AND 7

Appendix 5: APA PsycInfo

1. exp intensive care/ or "critical care".ti,ab. or "critically ill".ti,ab. or "critical illnesses".ti,ab. or "critical ill".ti,ab. or "critical illness".ti,ab. or "intensive care unit".ti,ab.
2. ((exp "Depression(emotion)"/ or exp Major Depression/ or exp caregiver burden/ or exp Hospital Admission/ or exp chronic pain/ or exp quality of life/ or exp Stress/) and Trauma Related Disorders/) or exp anxiety disorders/ or readmission.ti,ab. or "subsequent hospital admission".ti,ab. or readmitted.ti,ab. or readmit.ti,ab. or "patient readmission".ti,ab. or "depressive disorder".ti,ab. or "caregiver burden".ti,ab. or QoL.ti,ab. or "quality of life".ti,ab. or "chronic pain".ti,ab. or "intractable pain".ti,ab. or PTSD.ti,ab. or "post traumatic stress disorders".ti,ab. or "anxiety disorders".ti,ab.
3. 1 and 2
4. (random* or control*).ti,ab. or exp Randomized Controlled Trials/
5. 3 and 4

Appendix 6: ICTRP search strategy

1. Condition: ("critical care" OR "critically ill" OR "critical illnesses" OR "critical ill" OR "critical illness" OR "intensive care unit")
2. Intervention: ("readmission" OR "subsequent hospital admission" OR "readmitted" OR "readmit" OR "patient readmission" OR "depressive disorder" OR "caregiver burden" OR "QoL" OR

“quality of life” OR “chronic pain” OR “intractable pain” OR “PTSD” OR “post traumatic stress disorders” OR “anxiety disorders”)

3. 1 AND 2

Recruitment status is ALL.

Appendix 7: ClinicalTrials.gov search strategy

Condition or disease:

"critical care" OR "critically ill" OR "critical illnesses" OR "critical ill" OR "critical illness" OR "intensive care unit"

Study type:

Interventional Studies

Outcomes:

"Readmission" OR "subsequent hospital admission" OR "readmitted" OR "readmit" OR "depressive disorder" OR "caregiver burden" OR "QoL" OR "quality of life" OR "chronic pain" OR "intractable pain" OR "PTSD" OR "anxiety disorders"